Solid Geometry & Similarity

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Volume of prisms and cylinders

Find the volume of a cylinder with radius 5 cm and height 14 cm, in terms of π.

$V = \pi r^2 h = \pi(25)(14) = 350\pi$ cm^3.

2 [Foundational] • Target: Volume of prisms and cylinders

A triangular prism has a triangular cross-section of area 24 cm^2 and length 15 cm. Find its volume.

$V = (\text{cross} - \text{section area}) \times \text{length} = 24 \times 15 = 360$ cm^3.

3 [Foundational] • Target: Volume of prisms and cylinders

Find the volume of a cube with side length 9.

$V = 9^3 = 729$ cubic units.

4 [Foundational] • Target: Volume of pyramids and cones

Find the volume of a square pyramid with base side 10 and height 12.

$V = \frac{1}{3}(\text{base area})(\text{height}) = \frac{1}{3}(100)(12) = 400$ cubic units.

5 [Foundational] • Target: Volume of pyramids and cones

Find the volume of a cone with radius 6 and height 15, in terms of π.

$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(36)(15) = 180\pi$ cubic units.

6 [Foundational] • Target: Volume of pyramids and cones

A cone has volume 300π cubic units and height 25. Find its radius.

$300\pi = \frac{1}{3}\pi r^2(25) \Rightarrow r^2 = \frac{900}{25} = 36 \Rightarrow r = 6$.

7 [Foundational] • Target: Volume of pyramids and cones

A rectangular pyramid has a base 6 by 8 and height 9. Find its volume.

$V = \frac{1}{3}(6 \times 8)(9) = \frac{1}{3}(48)(9) = 144$ cubic units.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Volume of spheres

Find the volume of a sphere with radius 9, in terms of π.

$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(729) = 972\pi$ cubic units.

9 [Intermediate] • Target: Volume of spheres

A hemisphere has radius 6. Find its volume, in terms of π.

$V = \frac{1}{2}\left(\frac{4}{3}\pi r^3\right) = \frac{2}{3}\pi(216) = 144\pi$ cubic units.

10 [Intermediate] • Target: Volume of spheres

A spherical tank has diameter 10. Find its volume, in terms of π.

Radius $= 5$. $V = \frac{4}{3}\pi(125) = \frac{500\pi}{3}$ cubic units.

11 [Intermediate] • Target: Volume of spheres

Find the volume and surface area of a sphere with radius 3, in terms of π.

$V = \frac{4}{3}\pi(27) = 36\pi$ cubic units. $SA = 4\pi(9) = 36\pi$ square units.

12 [Intermediate] • Target: Surface area of solids

Find the surface area of a cylinder with radius 4 and height 10, in terms of π.

$SA = 2\pi r^2 + 2\pi rh = 2\pi(16) + 2\pi(4)(10) = 32\pi + 80\pi = 112\pi$.

13 [Intermediate] • Target: Surface area of solids

Find the surface area of a cone with radius 5 and slant height 13, in terms of π.

$SA = \pi r^2 + \pi rl = 25\pi + 65\pi = 90\pi$.

14 [Intermediate] • Target: Surface area of solids

Find the surface area of a sphere with radius 7, in terms of π.

$SA = 4\pi r^2 = 4\pi(49) = 196\pi$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Surface area of solids • Cross-Lesson: 8.01 Statistics & Probability

Find the surface area of a cube with side length 7.

$SA = 6(7^2) = 6(49) = 294$ square units.

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16 [Challenging] • Target: Composite solids • Cross-Lesson: 1.01 Advanced Algebra

A solid consists of a cylinder (radius 3, height 10) topped by a cone with the same radius and height 4. Find the total volume, in terms of π.

Cylinder: $\pi(9)(10) = 90\pi$. Cone: $\frac{1}{3}\pi(9)(4) = 12\pi$. Total $= 90\pi + 12\pi = 102\pi$.

17 [Challenging] • Target: Composite solids • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

A rectangular box measuring 8 by 6 by 4 has a cylindrical hole (radius 2, drilled fully through the height of 4) removed. Find the remaining volume, in terms of π.

Box volume $= 8 \times 6 \times 4 = 192$. Cylinder removed $= \pi(4)(4) = 16\pi$. Remaining volume $= 192 - 16\pi$.

18 [Challenging] • Target: Composite solids • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

A solid consists of a cylinder (radius 4, height 8) topped by a hemisphere of the same radius. Find the total volume, in terms of π.

Cylinder: $\pi(16)(8) = 128\pi$. Hemisphere: $\frac{2}{3}\pi(64) = \frac{128\pi}{3}$. Total $= 128\pi + \frac{128\pi}{3} = \frac{512\pi}{3}$.

19 [Challenging] • Target: Composite solids • Cross-Lesson: 4.01 Lines, Triangles & Polygons

A solid is a rectangular box (10 by 6 by 5) with a square pyramid (base 6 by 6, height 4) carved out of the top. Find the remaining volume.

Box: $10 \times 6 \times 5 = 300$. Pyramid: $\frac{1}{3}(36)(4) = 48$. Remaining volume $= 300 - 48 = 252$ cubic units.

20 [Challenging] • Target: Similar solids • Cross-Lesson: 5.01 Trigonometry

Two similar cylinders have heights 6 and 9. Find the ratio of their volumes.

Scale factor $k = \frac{9}{6} = \frac{3}{2}$. Volume ratio $= k^3 = \frac{27}{8}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Volume of prisms and cylinders

Two similar solids have volume ratio $1 : 64$. If a side of the smaller solid is 11, find the corresponding side of the larger solid.

a. 27

b. 19

c. **44** ✓

d. 30

Linear ratio $= \sqrt[3]{1 : 64} = 1 : 4 \Rightarrow \text{side} = 11 \times \frac{4}{1} = 44$.

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22 [Foundational] • Target: Volume of pyramids and cones

A solid is formed by a cylinder of radius 3 and height 13 topped with a hemisphere of the same radius. Find the total volume. (In terms of π.)

a. 164π

b. 27π

c. 135π ✓

d. 51π

Cylinder $= \pi r^2 h = 117\pi$. Hemisphere $= \frac{2}{3}\pi r^3 = 18\pi$. Total $= 135\pi$.

23 [Foundational] • Target: Volume of spheres

Find the volume of a cylinder with radius 2 and height 14. (Leave your answer in terms of π.)

a. 78π

b. 61π

c. 56π ✓

d. 22π

$V = \pi r^2 h = \pi(2)^2(14) = 56\pi$.

24 [Foundational] • Target: Surface area of solids

A cone with height H and base radius R is inscribed within a sphere of radius 5. The cone's vertex is at the sphere's south pole, and its base is a horizontal circle on the sphere. If the height of the cone is H = 8, what is the volume of the cone?

a. $\frac{128\pi}{3}$ ✓

b. $\frac{256\pi}{3}$

c. 128π

d. 256π

Let the sphere have its center at the origin (0,0) in a 2D cross-section. The sphere's radius is 5, so its equation is $x^2 + y^2 = 25$. The cone's vertex is at the south pole, which is at (0, -5). The height of the cone is H = 8. This means the base of the cone is at a y-coordinate of -5 + 8 = 3. Let R be the radius of the cone's base. This radius R is the x-coordinate of the point where the base circle intersects the sphere. So, we substitute y = 3 into the sphere's equation: $R^2 + 3^2 = 25$. This gives.

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25 [Foundational] • Target: Composite solids

A cone has a height of 18 cm. A smaller cone is formed by slicing the top of the original cone with a plane parallel to the base. The smaller cone is similar to the original cone, and its height is 12 cm. If the radius of the base of the original cone is 9 cm, what is the radius of the base of the smaller cone?

a. 6 **cm** ✓

b. 5 cm

c. 4.5 cm

d. 3 cm

Let the height of the original cone be H and its radius be R. Let the height of the smaller cone be h and its radius be r. We are given that $H = 18$ cm, $R = 9$ cm, and $h = 12$ cm. Since the smaller cone is similar to the original cone, the ratio of their corresponding linear dimensions is constant. Therefore, the ratio of their heights is equal to the ratio of their radii: $\frac{h}{H} = \frac{r}{R}$ Substituting the given values: $\frac{12}{18} = \frac{r}{9}$ To solve for r, we can simplify the fracti.

26 [Foundational] • Target: Similar solids

A right circular cone has a height of 12 cm. A smaller cone is formed by slicing the top of the larger cone with a plane parallel to the base. The smaller cone is similar to the larger cone, and its height is 6 cm. If the radius of the base of the larger cone is 9 cm, what is the radius of the base of the smaller cone?

a. 4.5 **cm** ✓

b. 5 cm

c. 6 cm

d. 7.5 cm

Let the larger cone be denoted by C_1 and the smaller cone by C_2. The height of C_1 is $H_1 = 12$ cm and its radius is $R_1 = 9$ cm. The height of C_2 is $H_2 = 6$ cm. Since the smaller cone is similar to the larger cone, the ratio of their corresponding linear dimensions is constant. Therefore, the ratio of their heights is equal to the ratio of their radii. $\frac{R_2}{R_1} = \frac{H_2}{H_1}$ Substitute the given values: $\frac{R_2}{9\text{ cm}} = \frac{6\text{ cm}}{12\text{ cm}}$.

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27 [Foundational] • Target: Cross-sections of solids

In the diagram below, two similar triangles ABC and ADE are shown, with point D on segment AB and point E on segment AC. If AD = 4, DB = 8, and AE = 6, what is the length of EC?

a. **12** ✓

b. 10

c. 8

d. 9

The problem states that triangle ABC is similar to triangle ADE. This means that the ratios of corresponding sides are equal. We are given AD = 4, DB = 8, so AB = AD + DB = 4 + 8 = 12. We are also given AE = 6. We need to find the length of EC. Since the triangles are similar, we have the proportion: $\frac{AD}{AB} = \frac{AE}{AC}$ Substituting the given values: $\frac{4}{12} = \frac{6}{AC}$ Simplify the left side: $\frac{1}{3} = \frac{6}{AC}$ Cross-multiply to solve for AC: $AC = 3 \times 6 = 18$ We know that AC =.

28 [Foundational] • Target: A well-designed word problem: a grain silo

Find the volume of a cube with edge length 11.

a. **1331** ✓

b. 777

c. 814

d. 1001

Applying Solid Geometry & Similarity principles, $V = s^3 = 11^3 = 1331$. Therefore, the verified correct option is 1331.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Volume of prisms and cylinders

A cylinder has volume 1053π and radius 9. Find its height.

a. 2

b. **13** ✓

c. 7

d. 15

$V = \pi r^2 h \Rightarrow 1053\pi = \pi(9)^2 h \Rightarrow h = 13$.

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30 [Intermediate] • Target: Volume of pyramids and cones

A solid cone has a height of 24 cm. A smaller cone is cut from the top of the larger cone by a plane parallel to the base. The smaller cone has a height of 8 cm. If the radius of the base of the larger cone is 12 cm, what is the radius of the base of the smaller cone?

a. **4 cm** ✓

b. 6 cm

c. 8 cm

d. 5 cm

Let H be the height of the larger cone and R be the radius of its base. Let h be the height of the smaller cone and r be the radius of its base. We are given $H = 24$ cm, $h = 8$ cm, and $R = 12$ cm. Since the smaller cone is cut by a plane parallel to the base, the smaller cone is similar to the larger cone. This similarity implies that the ratio of their corresponding linear dimensions is constant. Specifically, the ratio of the heights is equal to the ratio of the radii: $\frac{h}{H} = \frac{r}{R}$ Subst.

31 [Intermediate] • Target: Volume of spheres

In the diagram below, chords AB and CD intersect at point P. If AP = 6, PB = 8, and CP = 4, find the length of PD.

a. **3** ✓

b. 4.5

c. 6

d. 12

The intersecting chords theorem states that when two chords intersect inside a circle, the product of the lengths of the segments of one chord is equal to the product of the lengths of the segments of the other chord. Let the chords be AB and CD, intersecting at point P. According to the theorem, we have: AP × PB = CP × PD We are given the following lengths: AP = 6 PB = 8 CP = 4 We need to find the length of PD. Substitute the given values into the equation: 6 × 8 = 4 × PD 48 = 4.

32 [Intermediate] • Target: Surface area of solids

Find the angle θ between $\mathbf{u} = \langle 1, 1\sqrt{3} \rangle$ and $\mathbf{v} = \langle 0, 3 \rangle$.

a. 30° ✓

b. 90°

c. 60°

d. 45°

$\mathbf{u} \cdot \mathbf{v} = 0 + 1(k+2)\sqrt{3}$. $||\mathbf{u}|| = 2k$, $||\mathbf{v}|| = k + 2$. $\cos\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = 30^\circ$.

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33 [Intermediate] • Target: Composite solids

Identify the Cartesian curve represented by $r = 6\cos\theta$.

a. Circle of radius 6 centered at $(0, 0)$

b. **Circle of radius 3 centered at $(3, 0)$** ✓

c. Cardioid with parameter $a = 3$

d. Ellipse with semi-major axis 6

Multiply by r: $r^2 = 6r\cos\theta \Rightarrow x^2 + y^2 - 6x = 0 \Rightarrow (x-3)^2 + y^2 = 9$. Center $(3, 0)$, radius 3.

34 [Intermediate] • Target: Similar solids

Find the principal cube root of $8i$ in polar form.

a. $2\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$

b. $8\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$

c. $2(\cos 0 + i\sin 0)$

d. $2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$ ✓

$8i = 8\text{cis}(\pi/2)$. Principal cube root: $\sqrt[3]{8}\,\text{cis}\left(\frac{\pi/2}{3}\right) = 2\text{cis}(\pi/6)$.

35 [Intermediate] • Target: Cross-sections of solids

$\Delta ABC \sim \Delta DEF$ with side ratio $AB : DE = 2 : 3$. If the area of ΔABC is 24, find the area of ΔDEF.

a. **54** ✓

b. 53

c. 14

d. 63

Areas scale with the square of the side ratio: $\left(\frac{3}{2}\right)^2 \times 24 = 54$.

36 [Intermediate] • Target: A well-designed word problem: a grain silo

Find the focus and directrix of the parabola $y^2 = 8x$.

a. **Focus $(2, 0)$, Directrix $x = -2$** ✓

b. Focus $(-2, 0)$, Directrix $x = 2$

c. Focus $(8, 0)$, Directrix $x = -8$

d. Focus $(0, 2)$, Directrix $y = -2$

$4p = 8 \Rightarrow p = 2$. Focus $(p, 0) = (2, 0)$, Directrix $x = -p = -2$.

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37 [Intermediate] • Target: Volume of prisms and cylinders

Convert $r = \frac{4}{1-2\cos\theta}$ to Cartesian form and identify its type.

a. Parabola: $y^2 = 4(2)x$

b. **Hyperbola:** $3x^2 - y^2 + 16x + 16 = 0$ ✓

c. Circle: $x^2 + y^2 = 4$

d. Ellipse: $3x^2 + y^2 = 16$

$e = 2 > 1$, so it is a hyperbola. In Cartesian: $3x^2 - y^2 + 16x + 16 = 0$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Volume of pyramids and cones • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Convert polar coordinates $(r, \theta) = (4, \frac{\pi}{3})$ to rectangular (x, y).

a. $(2, -2\sqrt{3})$

b. $(-2, 2\sqrt{3})$

c. $(2\sqrt{3}, 2)$

d. $(2, 2\sqrt{3})$ ✓

$x = 4\cos\frac{\pi}{3} = 2$, $y = 4\sin\frac{\pi}{3} = 2\sqrt{3}$.

39 [Challenging] • Target: Volume of spheres • Cross-Lesson: 4.01 Lines, Triangles & Polygons

A conic section has eccentricity $e = 0.20$. Identify this conic section.

a. Hyperbola

b. Circle

c. Parabola

d. **Ellipse** ✓

Since $0 < e < 1$, $e = 0.20$ represents an ellipse.

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40 [Challenging] • Target: Surface area of solids • Cross-Lesson: 5.01 Trigonometry

Find the equation of a sphere in $\mathbb{R}^3$ centered at $(1, -2, 2)$ with radius 3.

a. $x^2 + y^2 + z^2 = 9$

b. $(x-1)^2 + (y+2)^2 + (z-2)^2 = 3$

c. $(x+1)^2 + (y-2)^2 + (z+2)^2 = 9$

d. $(x-1)^2 + (y+2)^2 + (z-2)^2 = 9$ ✓

$(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2 = r^2$.

41 [Challenging] • Target: Composite solids • Cross-Lesson: 6.01 Circles & Coordinate Geometry

An L-shaped region is formed by removing a 3×9 rectangular corner from a 20×20 rectangle. Find the area of the remaining region.

a. **373** ✓

b. 364

c. 382

d. 379

Area $= (20)(20) - (3)(9) = 400 - 27 = 373$.

42 [Challenging] • Target: Similar solids • Cross-Lesson: 8.01 Statistics & Probability

Find the sum of the infinite series $8 + \frac{8}{2} + \frac{8}{4} + \cdots$.

a. **16** ✓

b. 20

c. Diverges

d. 8

$S = \frac{a}{1 - 1/2} = 2(8) = 16$.

43 [Challenging] • Target: Cross-sections of solids • Cross-Lesson: 1.01 Advanced Algebra

Two similar triangles have corresponding side lengths in ratio $4:6$. If the smaller triangle has area 8, what is the area of the larger triangle?

a. 10

b. 22

c. 14

d. **18** ✓

Area ratio is the square of the side ratio: $\left(\frac{6}{4}\right)^2 = \frac{36}{16}$. Larger area $= 8 \times \frac{36}{16} = 18$.

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44 **[Challenging]** • **Target: A well-designed word problem: a grain silo** • **Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic**

A survey was conducted on the preferred modes of transportation for students travelling to school. The results are summarized in the two-way table below. If a student is chosen at random from those surveyed, and it is known that they prefer to travel by bus, what is the probability that they are in Year 10? Let N_{10} represent the number of Year 10 students who prefer the bus, and N_{Bus} represent the total number of students who prefer the bus. The probability we are looking for is $\frac{N_{10}}{N_{Bus}}$. | Year Group | Bicycle | Bus | Train | Walk | Total | |---|---|---|---|---|---| | Year 9 | 45 | 60 | 25 | 70 | 200 | | Year 10 | 55 | x | 35 | 80 | 270 | | Year 11 | 30 | 50 | 40 | 50 | 170 | | Total | 130 | 160 | 100 | 200 | 640 |

a. $\frac{130}{160}$

b. $\frac{x}{160}$ ✓

c. $\frac{x}{270}$

d. $\frac{160}{640}$

The question asks for the conditional probability that a student is in Year 10, given that they prefer to travel by bus. This can be represented as $P(\text{Year 10}|\text{Bus})$. Using the formula for conditional probability, $P(A|B) = \frac{P(A \cap B)}{P(B)}$. In this context, A is the event that a student is in Year 10, and B is the event that a student prefers to travel by bus. The number of students in Year 10 who prefer the bus is given as x in the table. The total number of students who prefer.

45 **[Challenging]** • **Target: Volume of prisms and cylinders** • **Cross-Lesson: 3.01 Sequences, Series & Complex Numbers**

A circle has a radius of 6 units. A chord within this circle subtends a sector with a central angle of 60 degrees. What is the length of the chord?

a. **6 units** ✓

b. $6\sqrt{3}$ units

c. 12 units

d. $12\sqrt{3}$ units

We have a circle with radius $r = 6$ and a sector with a central angle of 60 degrees. The chord connecting the endpoints of the arc of this sector forms an isosceles triangle with the two radii. Since the central angle is 60 degrees and the other two angles in the triangle must be equal (because the sides opposite them are radii), each of the other angles is $(180^\circ - 60^\circ)/2 = 60^\circ$. Therefore, the triangle is equilateral. In an equilateral triangle, all sides are equal. Since two sides are radii of.

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